Lecture 4: Demand and the Consumer

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Lecture overview

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- Marginal Utility Theory
- Demand and the Consumer The Characteristics Approach

Choice and Rationality

- **The Problem of Choice**: Limited incomes force consumers to make trade-offs (e.g., buying a textbook vs. a concert ticket).

- **Business Interest:** Managers study consumer decision-making to optimize pricing, packaging, and sales strategies.

- **Rational Behavior Assumption:** Economists assume consumers act rationally, weighing relative costs and benefits to maximize satisfaction from their limited budget.

The Concept of Utility

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- **Definition of Utility:** The satisfaction or pleasure derived from consuming a good or service.

- **Total Utility (TU):** The cumulative satisfaction gained from all units of a commodity consumed over a specific time (e.g., the total pleasure from 10 cups of tea).

- **Marginal Utility (MU):** The extra satisfaction gained from consuming one additional unit of a good.

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Marginal Utility Theory

Marginal Utility Theory

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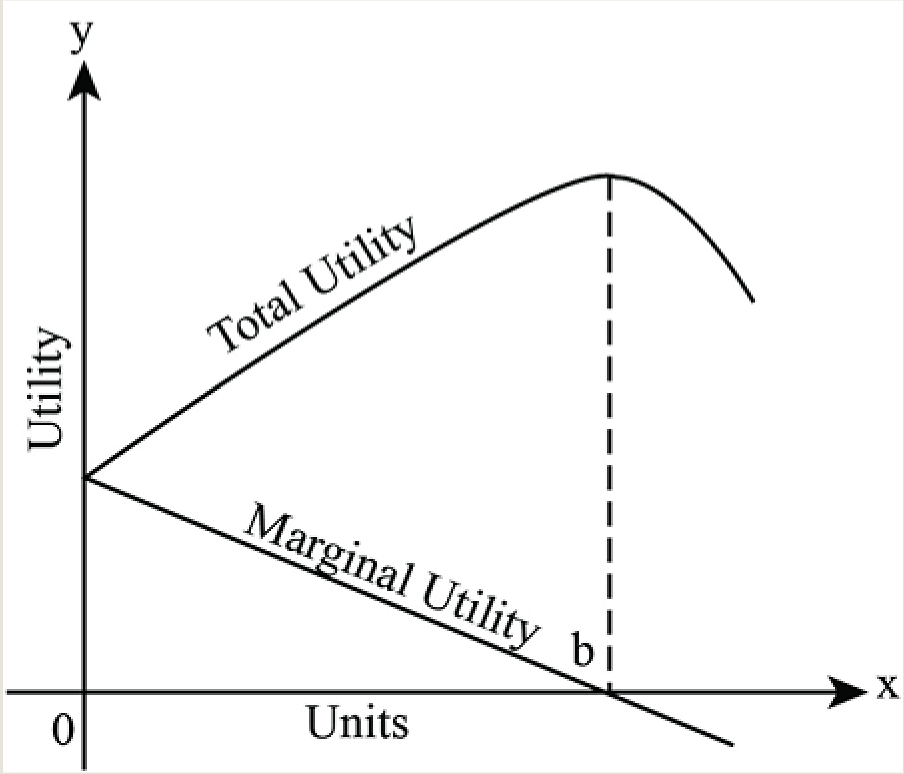
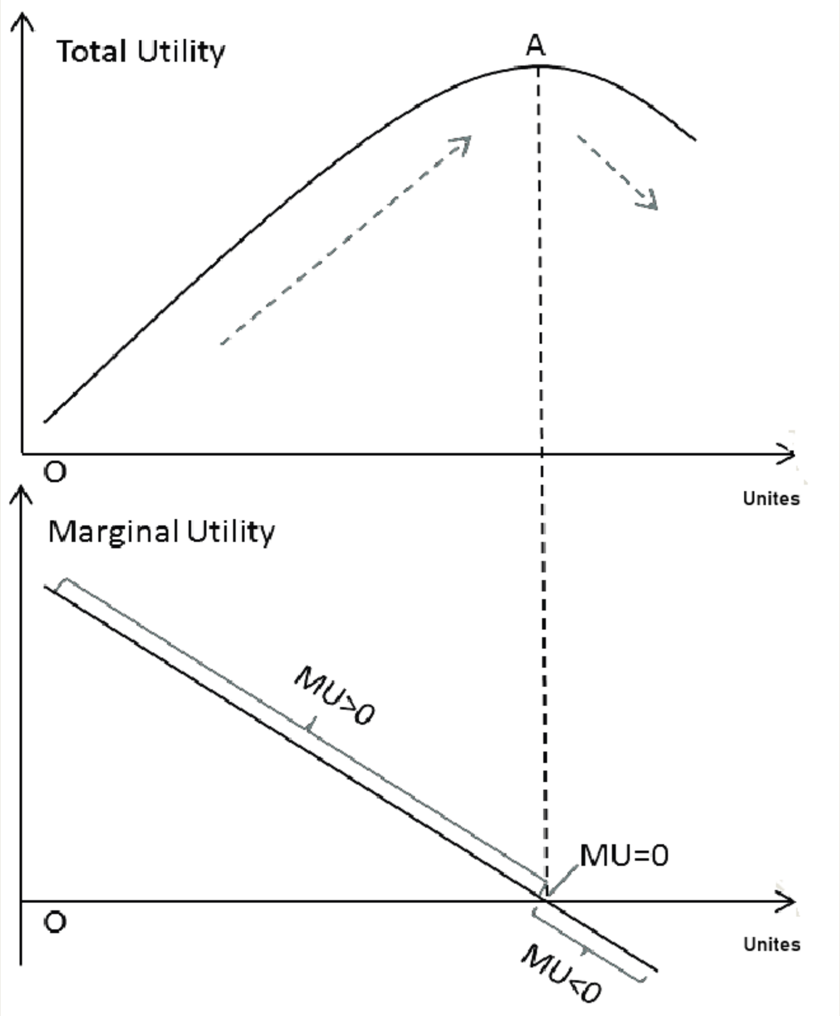
- **Principle of Diminishing Marginal Utility:** As consumption increases, the additional satisfaction (MU) from each extra unit decreases.
    - **Example:** The first cup of tea provides high satisfaction; the third cup provides much less.

Marginal Utility Theory





Marginal Utility Theory

- **Measuring Utility with Money:** Since utility is subjective, economists often use the maximum amount a person is willing to pay as a proxy for utility.
- If a packet of crisps is worth 70p to Darren, his MU = 70p

Utility from consuming crisps

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Packet of crisps	*TU* in utils	*MU* in utils
0	0	-
1	0.7	0.7
2	0.11	0.4
3	0.13	0.2
4	0.14	0.1
5	0.14	0
6	0.13	-0.1

Q Kathy eats five slices of pizza one evening but admits that each slice of pizza doesn't taste as good as the previous one. This suggests that for Kathy:

A. the *TU* from pizzas is past its maximum point.

B. the *MU* of a slice of pizza is negative.

C. the *MU* is still rising but by smaller amounts.

D. the *MU* is positive but declining.

E. the *TU* is positive but declining.

Q Kathy eats five slices of pizza one evening but admits that each slice of pizza doesn't taste as good as the previous one. This suggests that for Kathy:

A. the *TU* from pizzas is past its maximum point.

- That would mean TU is already decreasing—this would require MU to be negative. But the statement only says it tastes **less good**, not that it becomes bad or harmful. So A isn't implied.

B. the *MU* of a slice of pizza is negative.

- Again, negative MU would mean an additional slice makes her worse off. The question does not state that; it only says each additional slice is less enjoyable than the last.

Q Kathy eats five slices of pizza one evening but admits that each slice of pizza doesn’t taste as good as the previous one. This suggests that for Kathy:

C. the *MU* is still rising but by smaller amounts.

- That contradicts the wording. “Doesn’t taste as good” implies MU is declining, not rising.

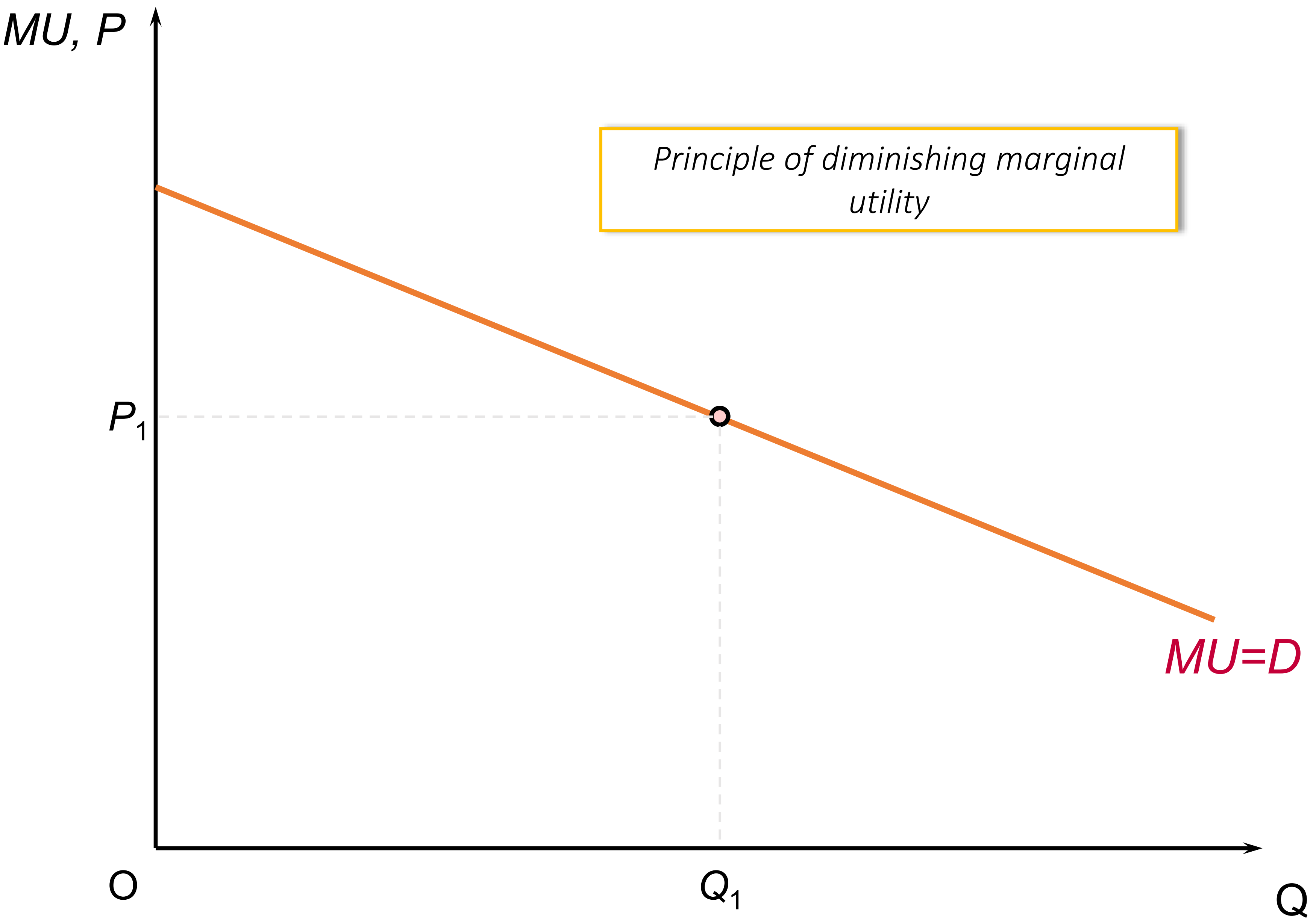
D. the *MU* is positive but declining.

- This matches the wording best: each additional slice still provides some satisfaction, but less than the previous one.

E. the *TU* is positive but declining.

- TU being positive just means overall satisfaction is above zero; TU “declining” would again imply MU negative beyond some point, which isn’t stated.

Consumer surplus



Marginal Utility Theory

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- **Marginal Consumer Surplus (MCS):** The difference between what you are willing to pay (MU) and the actual price (P).

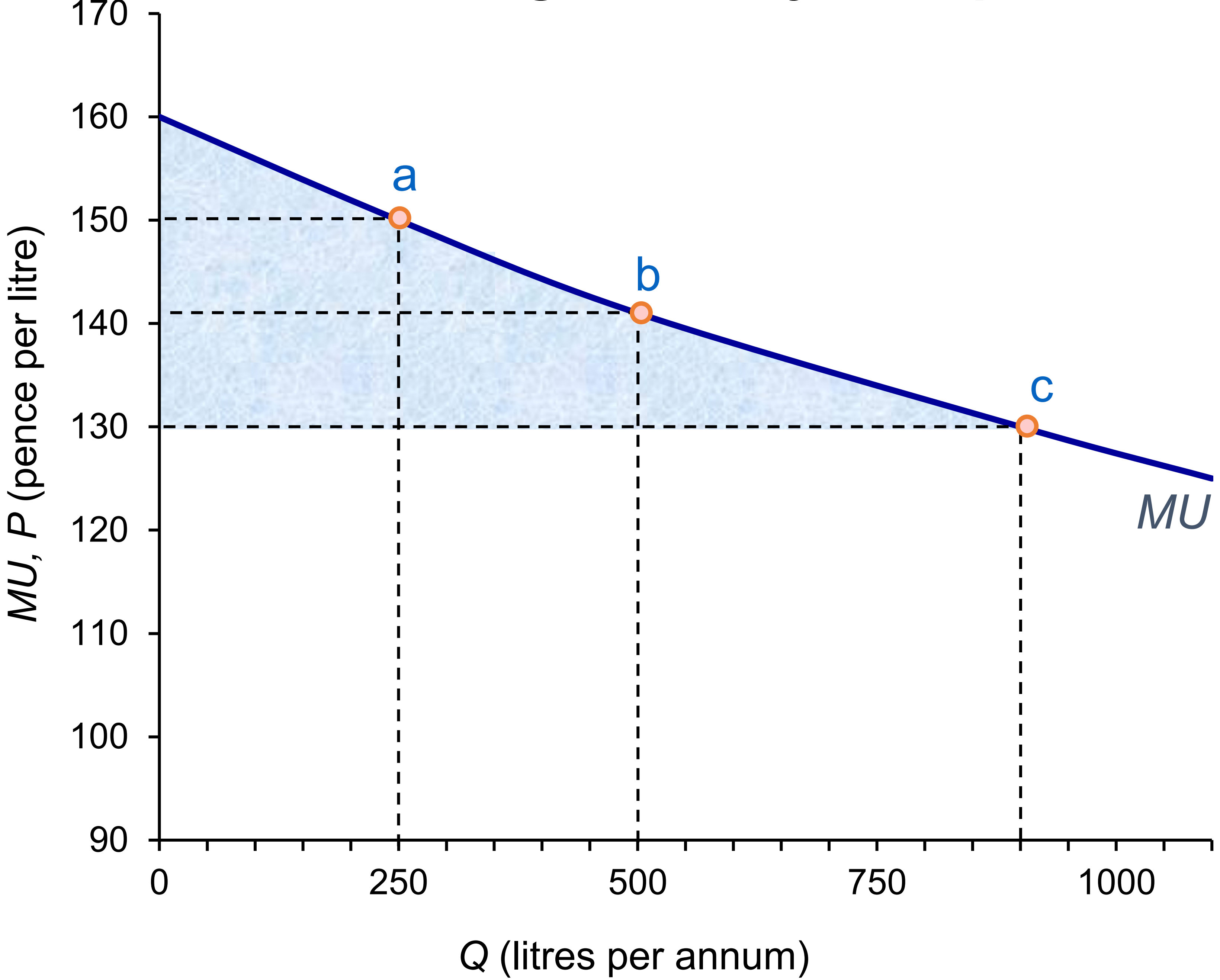
MCS = MU – P

  - Example: If Darren is willing to pay 70p but the price is 55p, his MCS = 15p.
- Total Consumer Surplus (TCS): The sum of all marginal consumer surpluses from all units consumed.

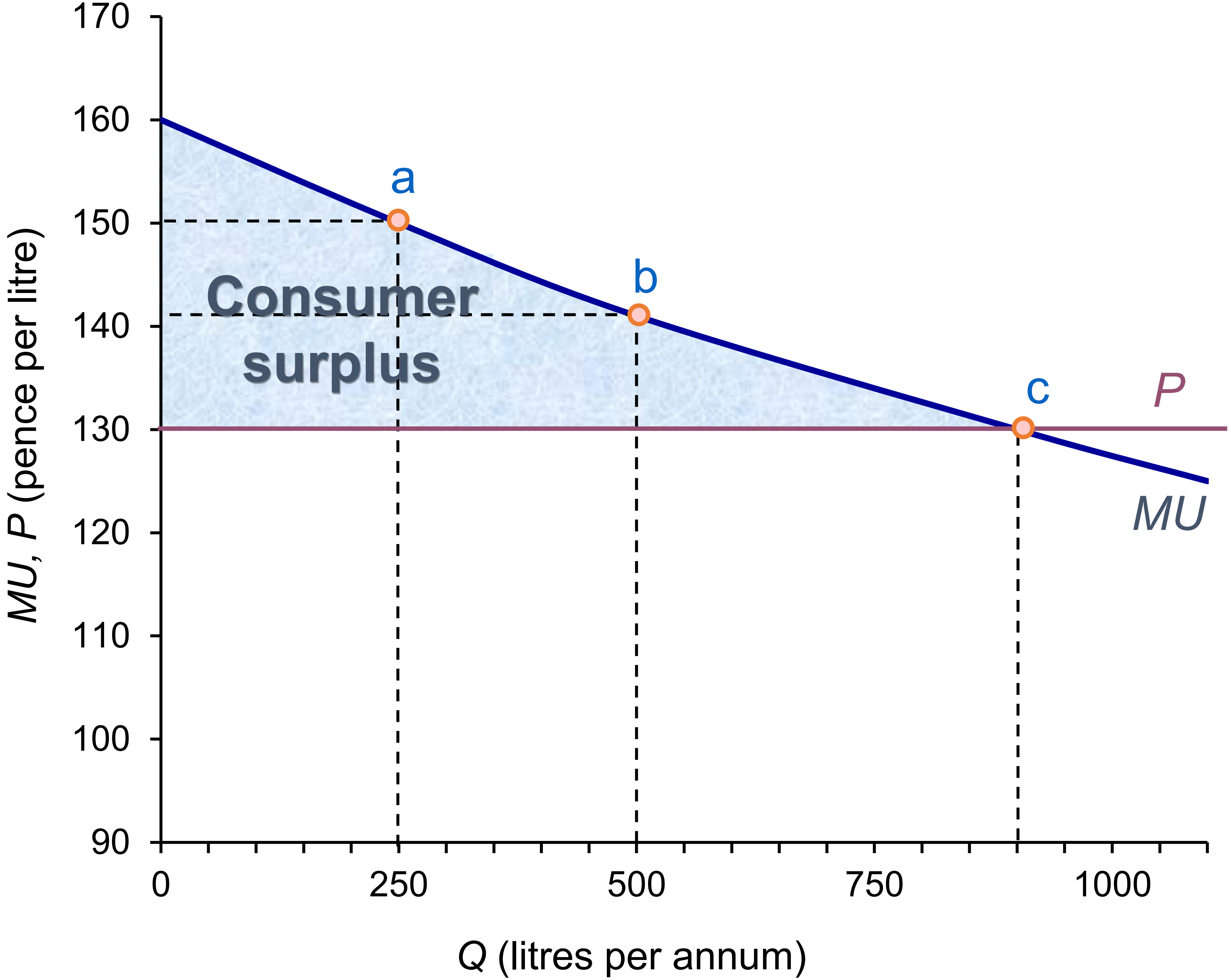
TCS = TU – TE

- Total Expenditure (TE): Calculated as $Price \times Quantity$ ($P \times Q$)
- Note: TE is the consumer's side of "Total Revenue" (TR).
- rational consumer behaviour
  - maximising consumer surplus: P = MU

Tina's marginal utility from petrol



Tina's consumer surplus from petrol



Deriving the Individual Demand Curve

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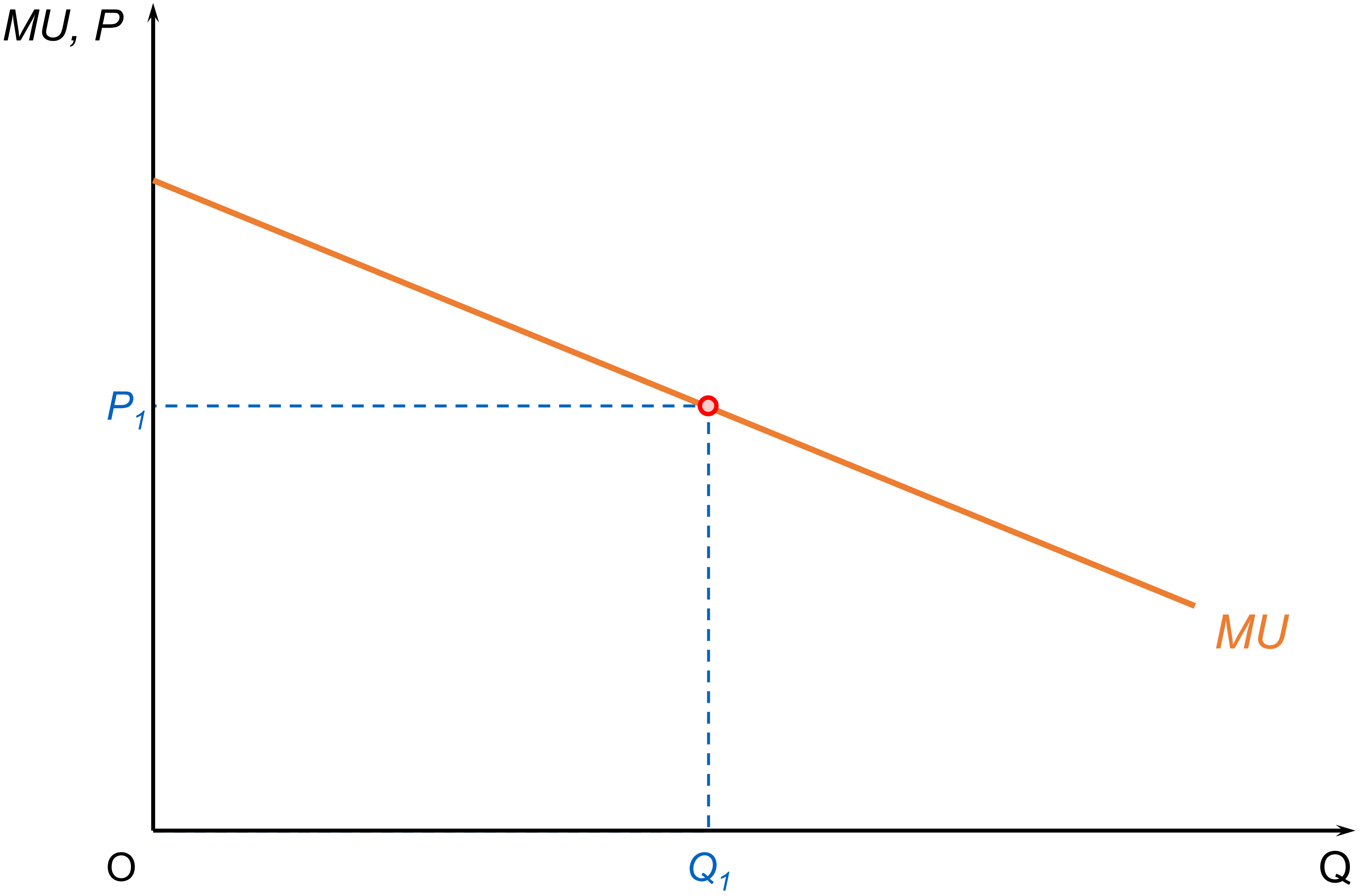
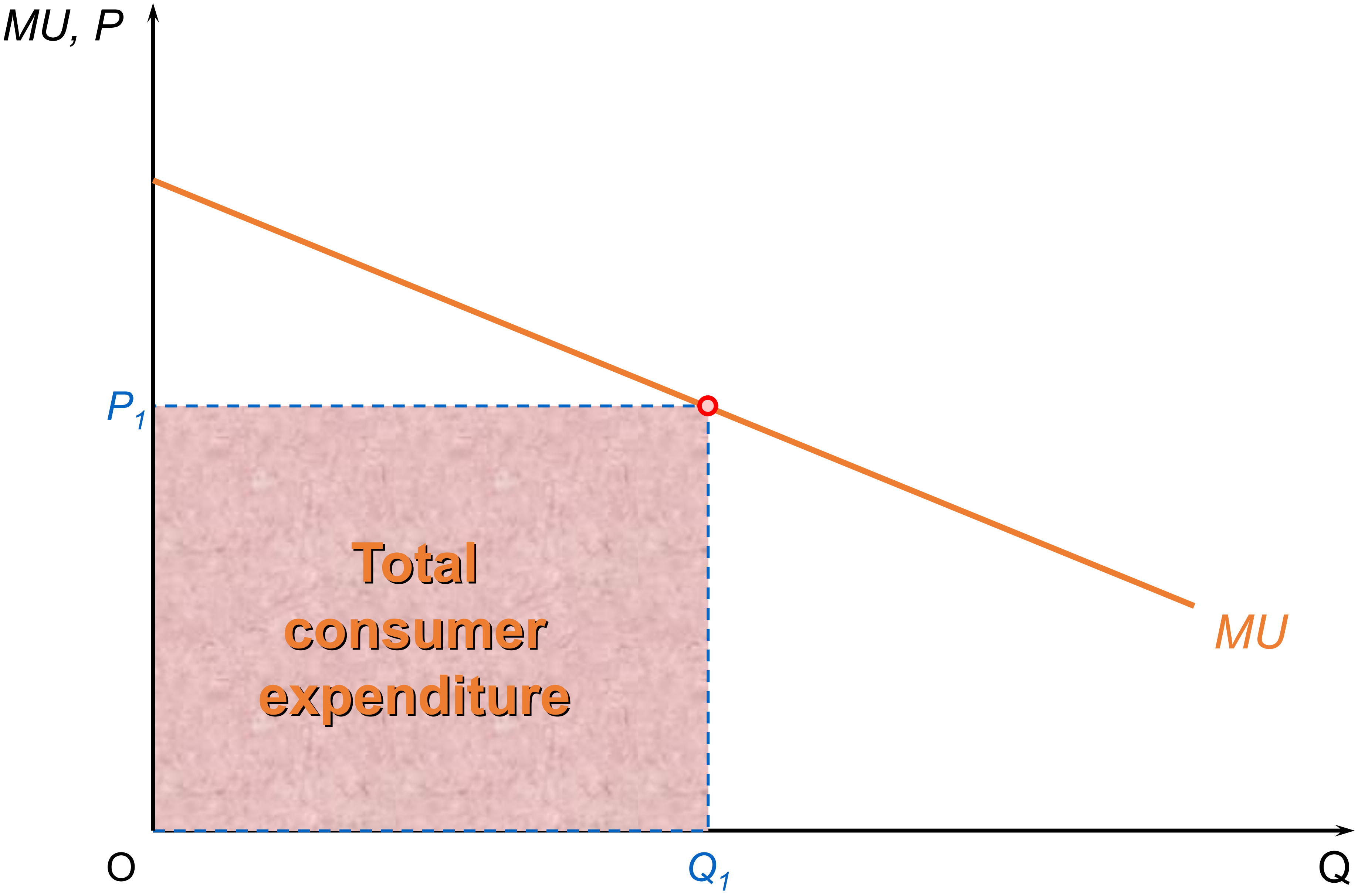
An individual's demand curve is identical to their Marginal Utility (MU) curve (when MU is measured in money).

- At price P1, the consumer chooses quantity Q1 to ensure MU = P1(Point a).
- If the price falls to P2, the consumer must increase consumption to Q2 so that MU falls to meet the new, lower price (Point b).
- If the price falls further to P3, they consume Q3 (Point c).
- Because consumers always aim to consume where P = MU, every point on the MU curve represents a point on the demand curve. The downward slope of the demand curve is a direct result of diminishing marginal utility.

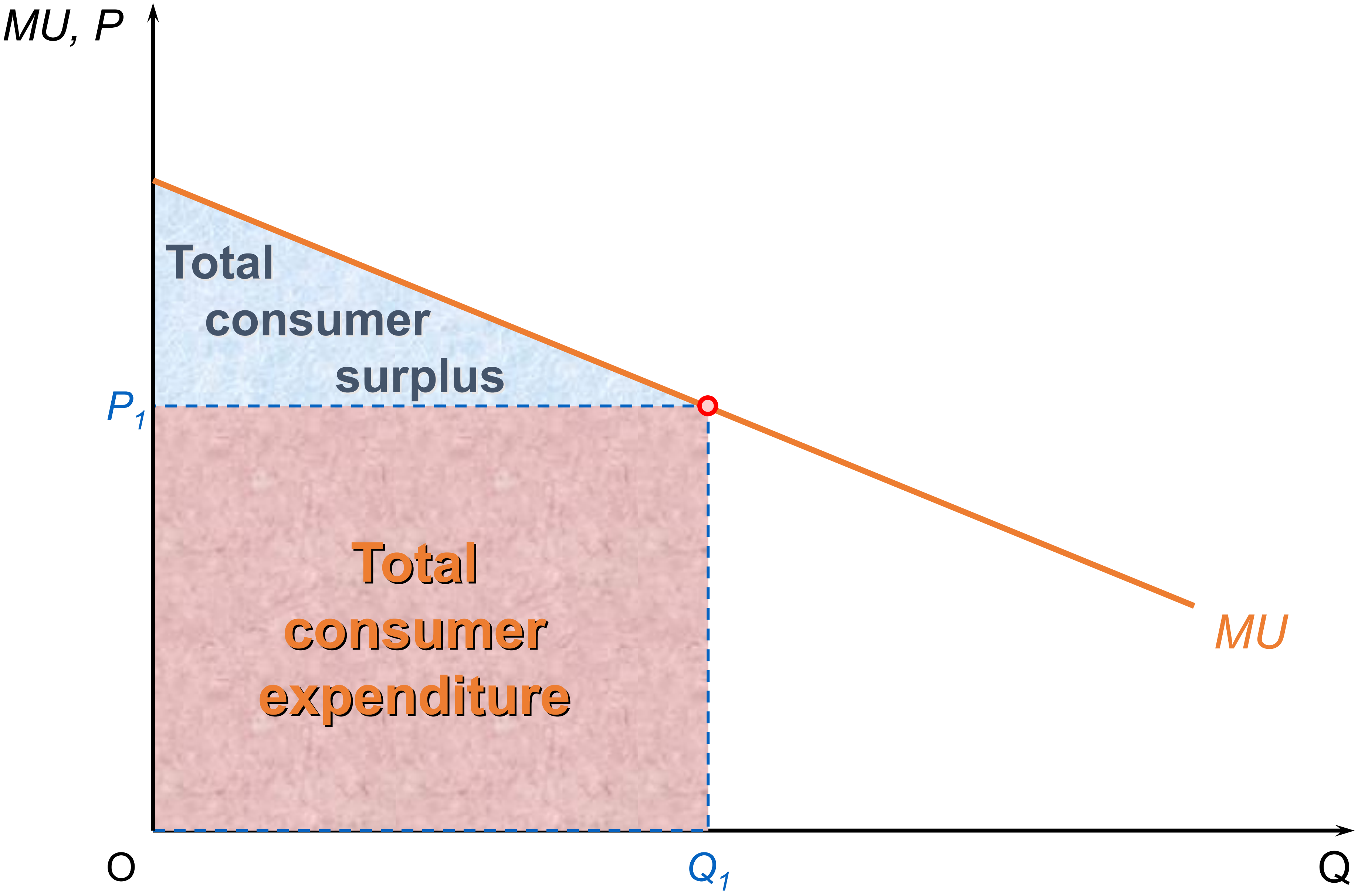
Consumer surplus



Consumer surplus



Consumer surplus



Q Rational consumer behaviour is where a person consumes the amount of a good that:

A. maximises the total utility from the good.

B. maximises the consumer surplus from the good.

C. minimises the amount spent on the good to achieve a given level of utility.

D. maximises the marginal utility from the good.

E. equates the marginal utility with that from other goods.

Q Rational consumer behaviour is where a person consumes the amount of a good that:

A. maximises the total utility from the good.

- TU is maximised where MU = 0, but that ignores price. Rational choice with a price is not simply maximise TU; it's choose the best given the cost.

B. maximises the consumer surplus from the good.

C. minimises the amount spent on the good to achieve a given level of utility.

- That's a different optimisation framing (cost-minimisation for a target).

Q Rational consumer behaviour is where a person consumes the amount of a good that:

D. maximises the marginal utility from the good.

- That would typically point to consuming only the first unit, since MU declines with quantity. That’s not the rational rule here.

E. equates the marginal utility with that from other goods.

- That’s a multi-good allocation condition

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Demand and the Consumer The Characteristics Approach

Choosing Between Products

- Consumers do not look at products in isolation; they compare them against rivals based on both price and product characteristics.
- Non-Price Factors: When buying a complex good (like a car), consumers evaluate a bundle of features:
    - Performance, comfort, and safety.
    - Reliability, durability, and fuel economy.
    - Added features (e.g., air conditioning, tech systems).
- Business Strategy: Firms constantly redesign products to maximize appeal relative to manufacturing costs, aiming to offer a more attractive "bundle" than competitors.

Characteristics Theory (Attributes Theory)

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- Utility is derived from the characteristics a product possesses, not just the product itself.
- Four Key Assumptions:

1. **Possession**: All products contain various distinct characteristics.
2. **Proportions**: Different brands offer these characteristics in different ratios.
3. **Objectivity**: Characteristics are measurable and objective (e.g., grams of fiber, horsepower).
4. **Choice Determinants:** Consumer choices are dictated by these characteristics in combination with price and income.

Plotting Characteristics (The Two-Attribute Model)

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- Imagine a market for breakfast cereal where consumers only care about two attributes: Taste (Sugar) and Health (Fiber).
- The Characteristics Diagram:
    - **Vertical Axis:** Quantity of Fiber.
    - **Horizontal Axis**: Quantity of Sugar.
    - **Product Rays**: Each brand is represented by a "ray" (a line) starting from the origin.
    - **The Slope:** The slope of the ray represents the ratio of fiber to sugar.
    - **Healthbran**: A steep ray (high fiber, low sugar).
    - **Tastyflakes**: A shallow ray (high sugar, low fiber).

Plotting Characteristics (The Two-Attribute Model)

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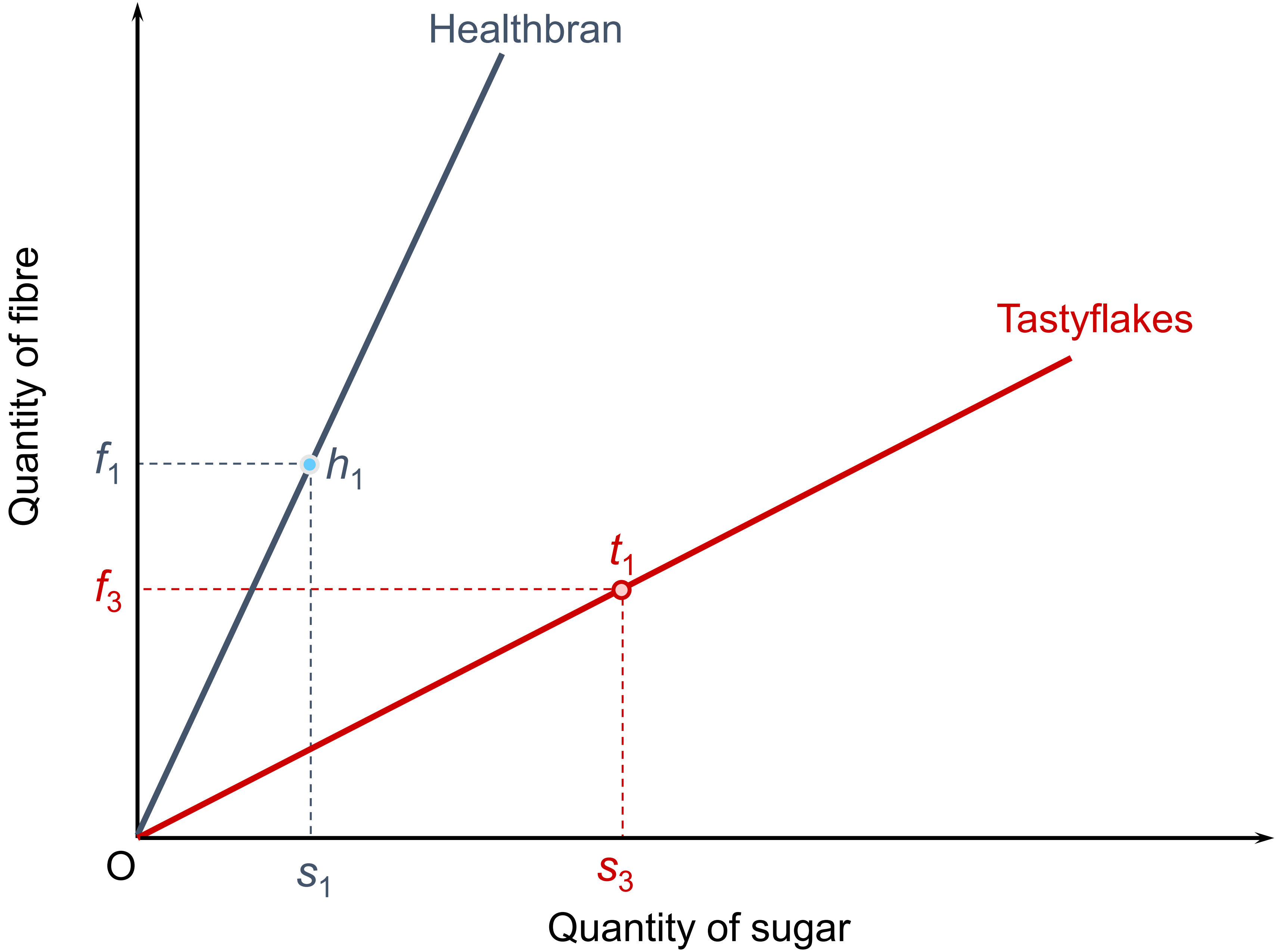
- **Consumption Points:** Moving further out along a ray means the consumer is buying more of that brand, increasing the total amount of both attributes while keeping the ratio constant.

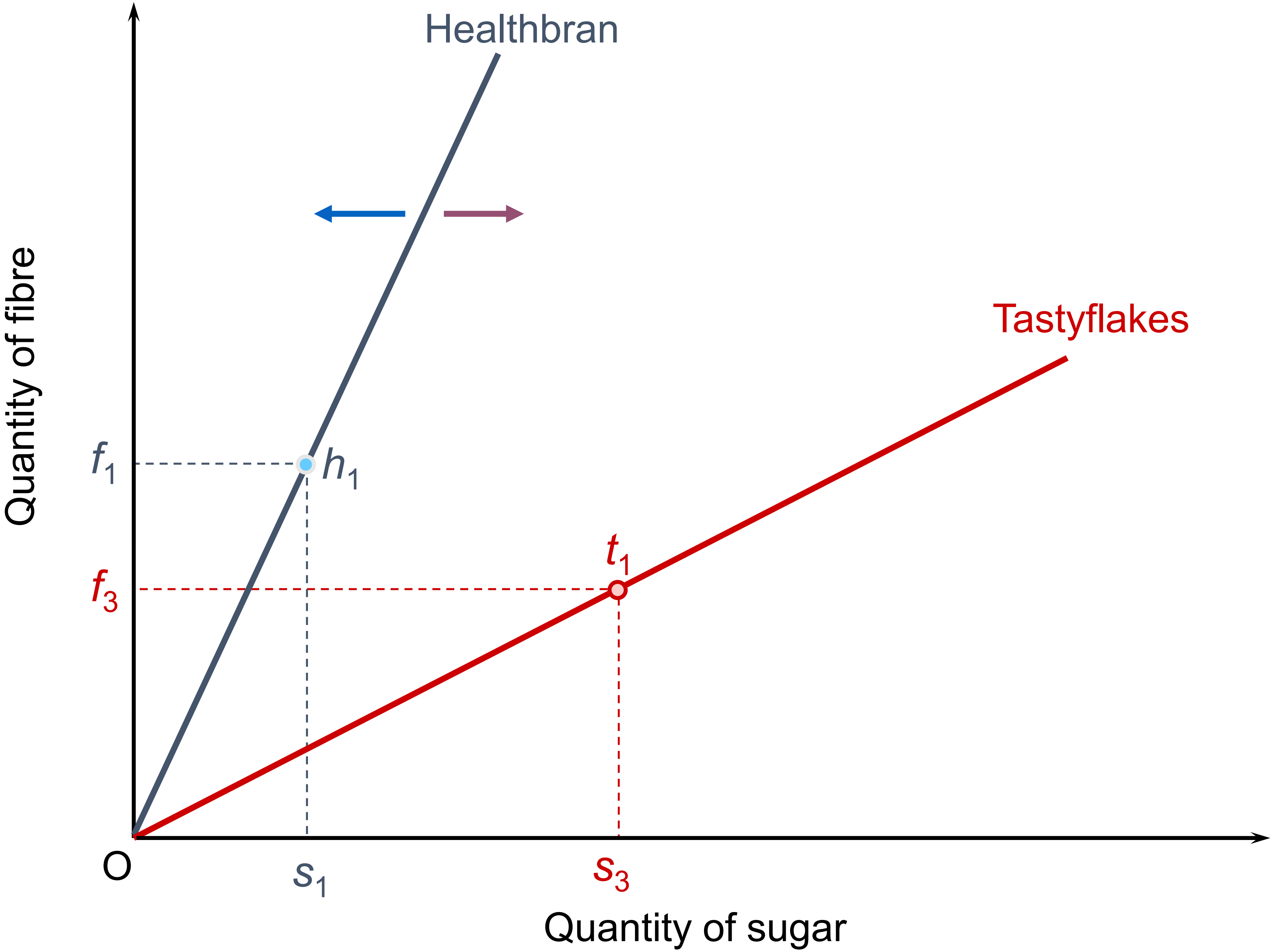
The characteristics of two brands of breakfast cereal



Changes in Product Characteristics

- Product Reformulation: If a firm changes its recipe or design, the "mix" of attributes changes.
- Visual Impact: A change in the attribute mix causes the slope of the ray to shift.
    - Example: If Healthbran is reformulated to be sweeter, its ray moves toward the sugar axis, becoming shallower.
    - Example: If Tastyflakes increases its fiber content, its ray moves toward the fiber axis, becoming steeper.

The characteristics of two brands of breakfast cereal



The Budget Constraint on Attributes

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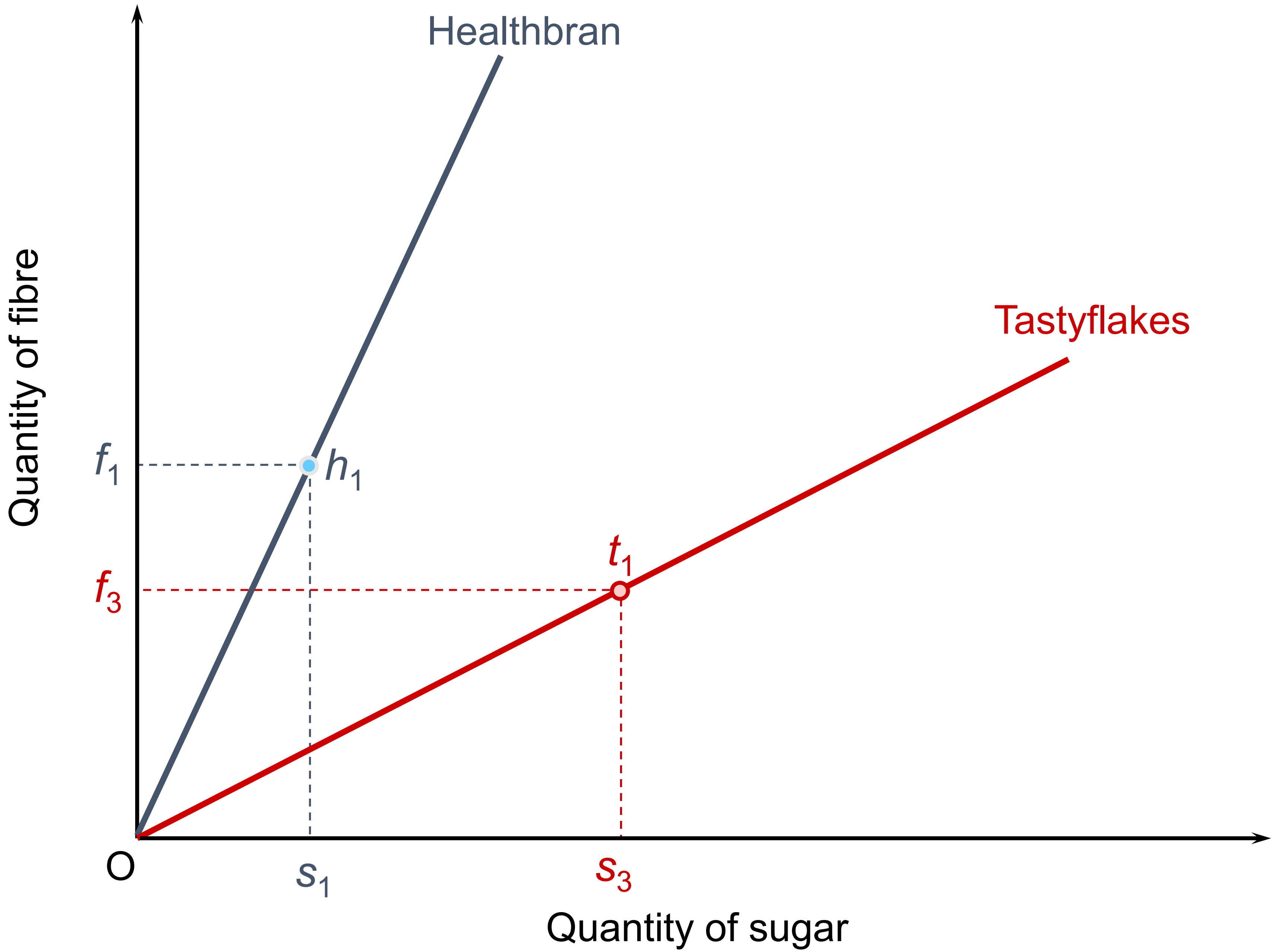
- A consumer’s ability to acquire characteristics is limited by their budget and the price of the brand.
- If Jane allocates a specific budget to breakfast cereal and chooses only one brand (Healthbran), her maximum consumption is represented by a single point (e.g., h1) on that brand's ray.
- At point h1, she receives a specific quantity of fiber f1 and sugar s1.

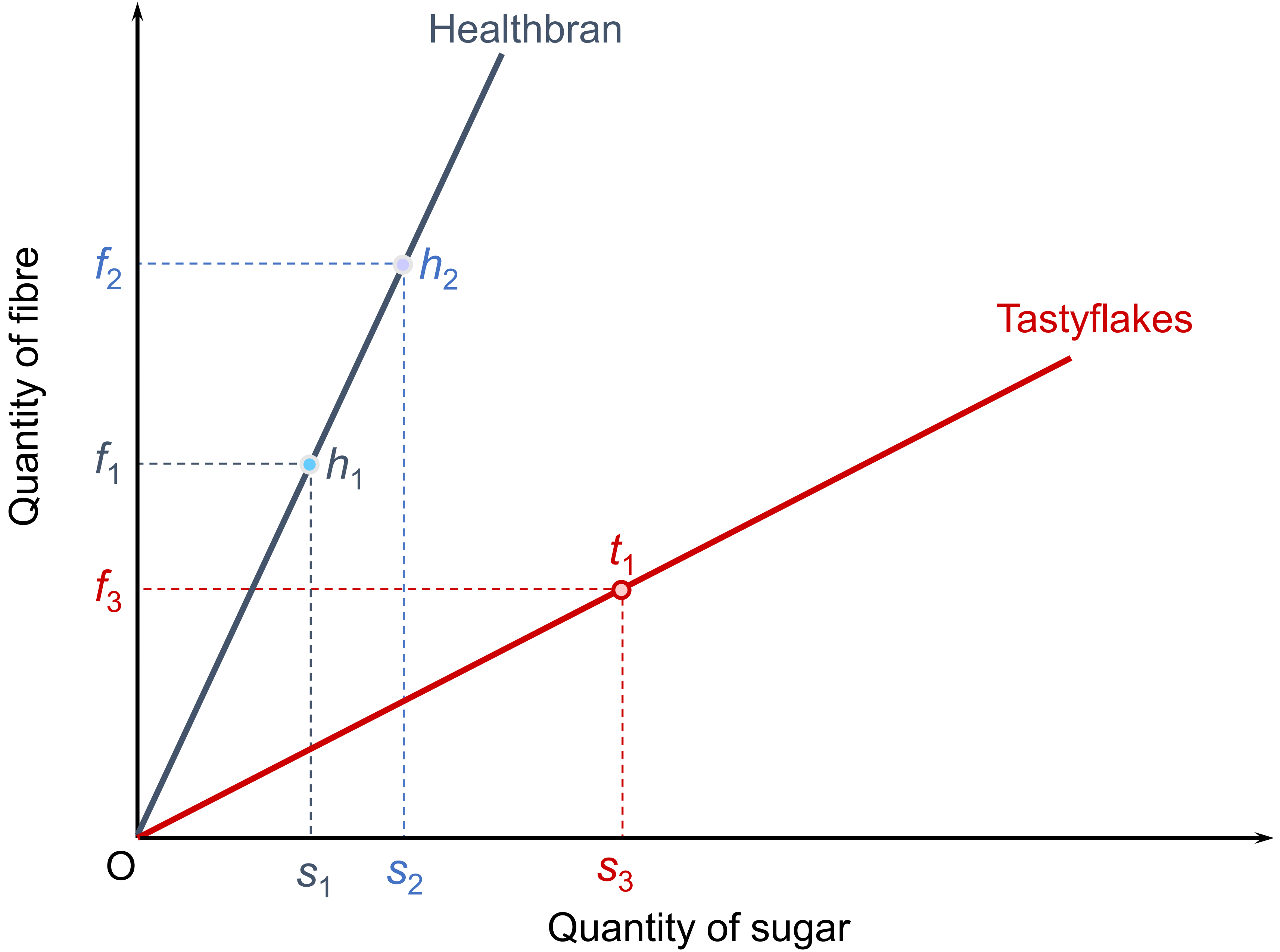
The characteristics of two brands of breakfast cereal



Impact of Budget Changes

- If Jane allocates more income to cereal while sticking to the same brand, she moves up the ray (e.g., to h2).
- This results in more of both attributes (fiber and sugar).
- Conversely, a reduction in the allocated budget results in a movement down the ray, reducing the total amount of characteristics obtained.

The characteristics of two brands of breakfast cereal



Impact of Price Changes

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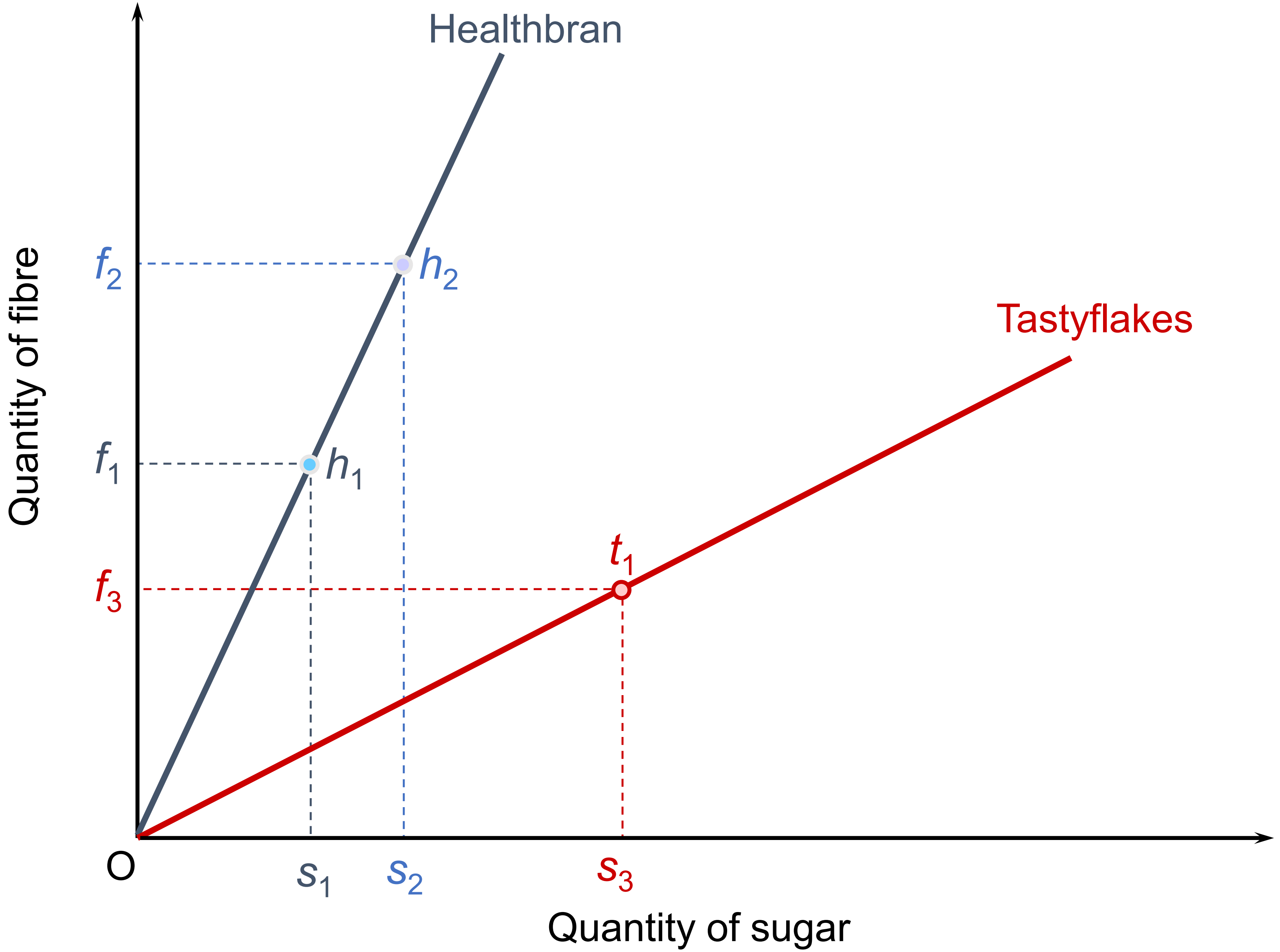
- If the price of a brand rises but the consumer's budget remains the same, their purchasing power decreases.
- Visual Movement: The consumer is forced to move down the ray because their fixed budget now buys fewer units of the product, and thus fewer characteristics.

The characteristics of two brands of breakfast cereal



The Efficiency Frontier (Mixing Brands)

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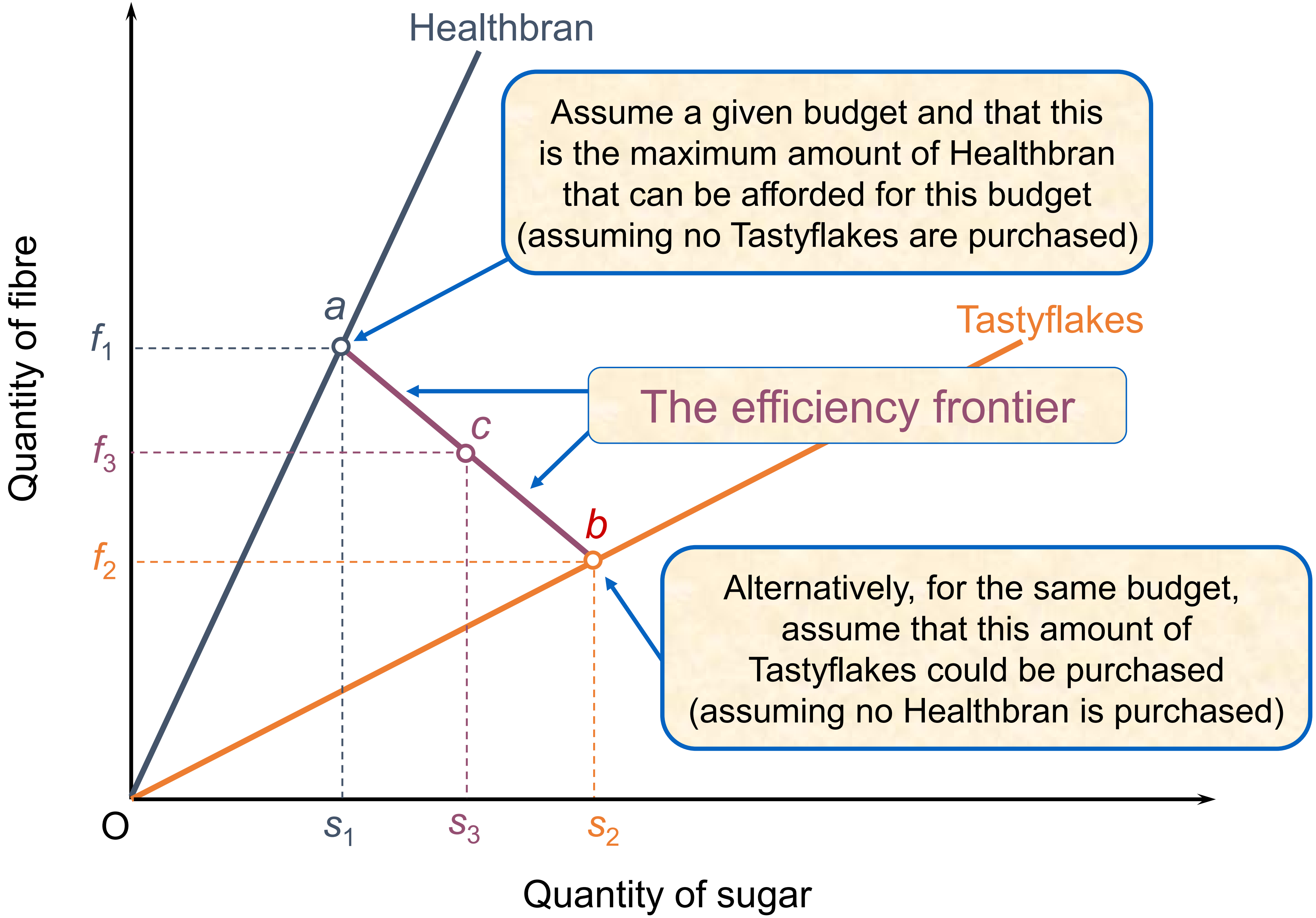
- Consumers often buy a variety of brands to avoid diminishing marginal utility from a single characteristic mix.
- If Jane can afford point a (Healthbran) or point b (Tastyflakes), she can also choose any combination of the two.
- **The Efficiency Frontier:** This is the straight line connecting points a and b. It represents the maximum combinations of attributes available to the consumer for a fixed budget.
    - Example: By spending half her budget on each, she might reach point c, achieving a "middle ground" of fiber and sugar.

The efficiency frontier



Multi-Brand Frontiers and Shifts

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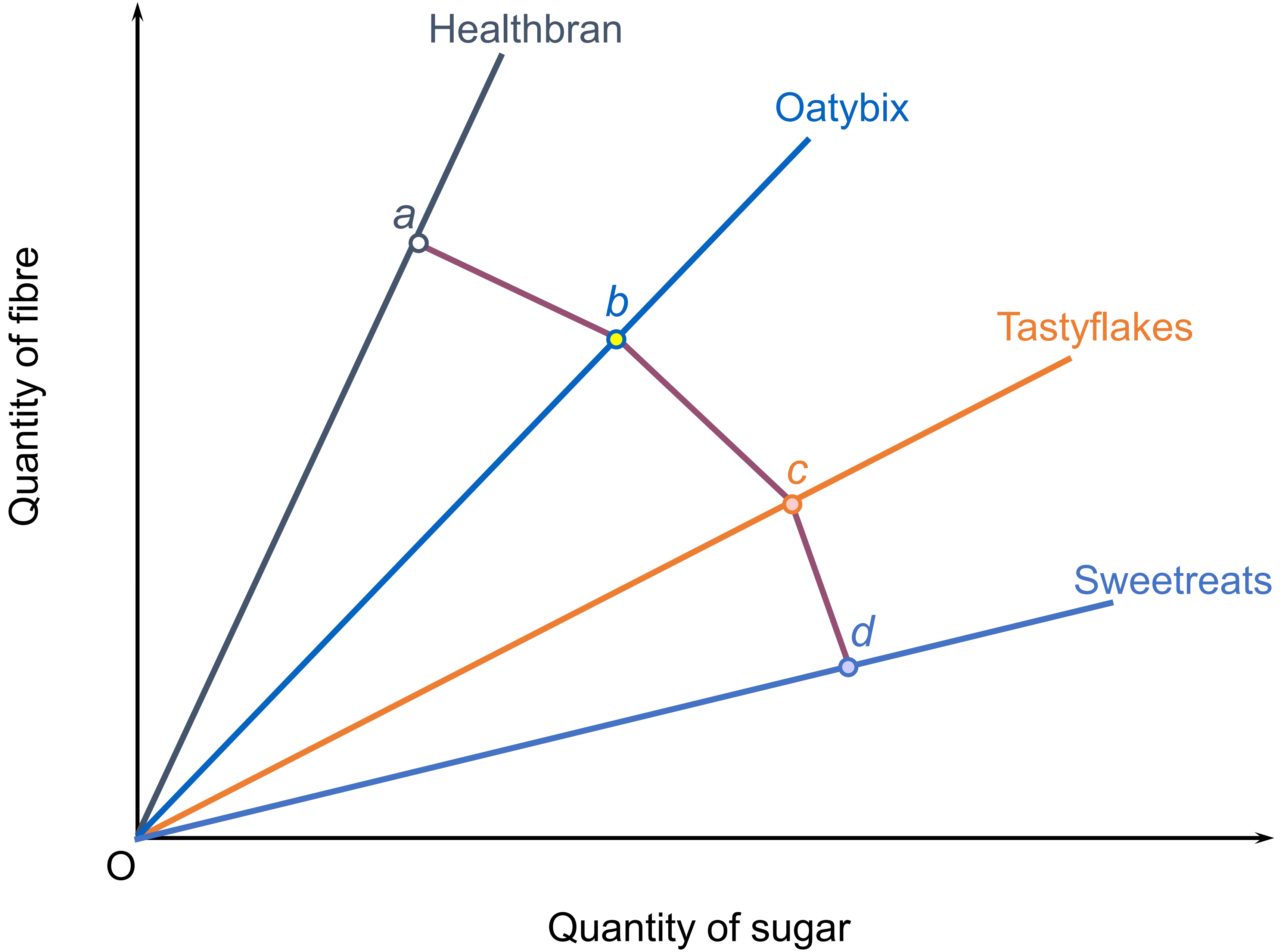
- Complex Frontiers: With multiple brands (e.g., four cereals), the efficiency frontier becomes a series of connected segments, forming a "boundary" of the best possible attribute combinations.
- If one brand (e.g., Oatybix) becomes more expensive, its point on the ray moves inward, pulling that section of the efficiency frontier toward the origin.
- If a brand changes its recipe, its ray pivots (changes slope), altering the frontier's shape.
- If the total budget increases or decreases, the entire efficiency frontier shifts parallel (outward or inward) across all rays.

The efficiency frontier: four brands



Q In the diagram, if brand X became more expensive, the efficiency frontier would:

A. shift outwards parallel to its original position.

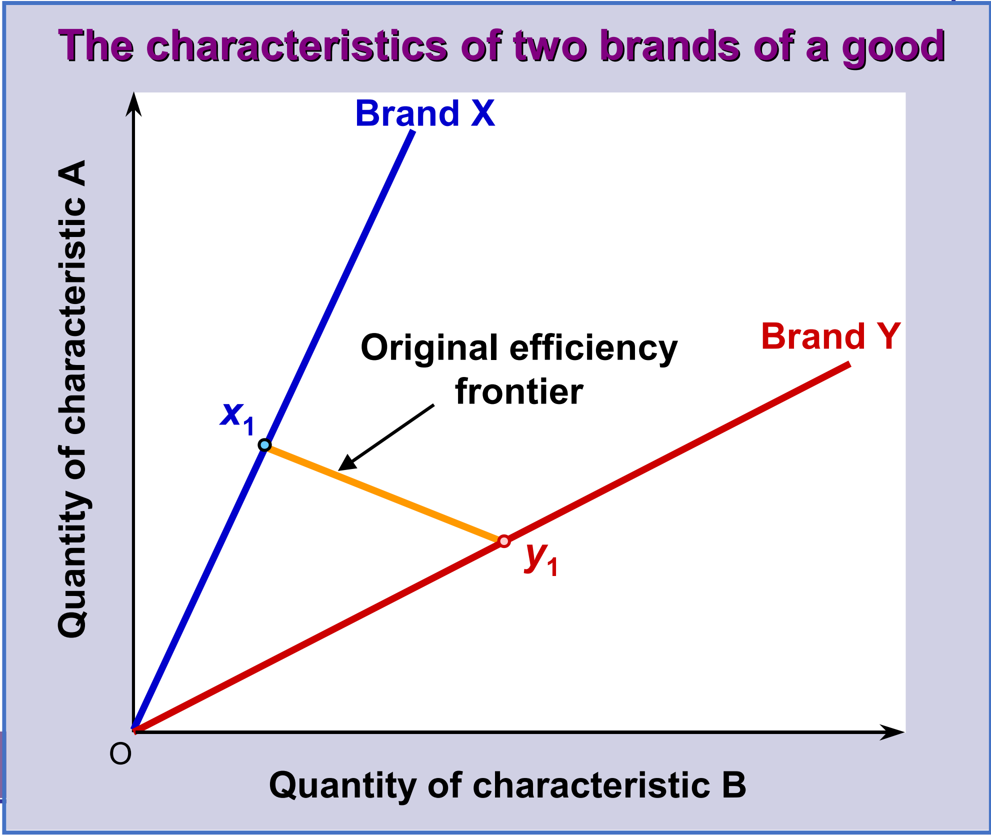
B. shift inwards parallel to its original position.

C. become steeper, passing through point y_1.

D. become flatter, passing through point y_1.

E. become steeper, passing through point x_1.

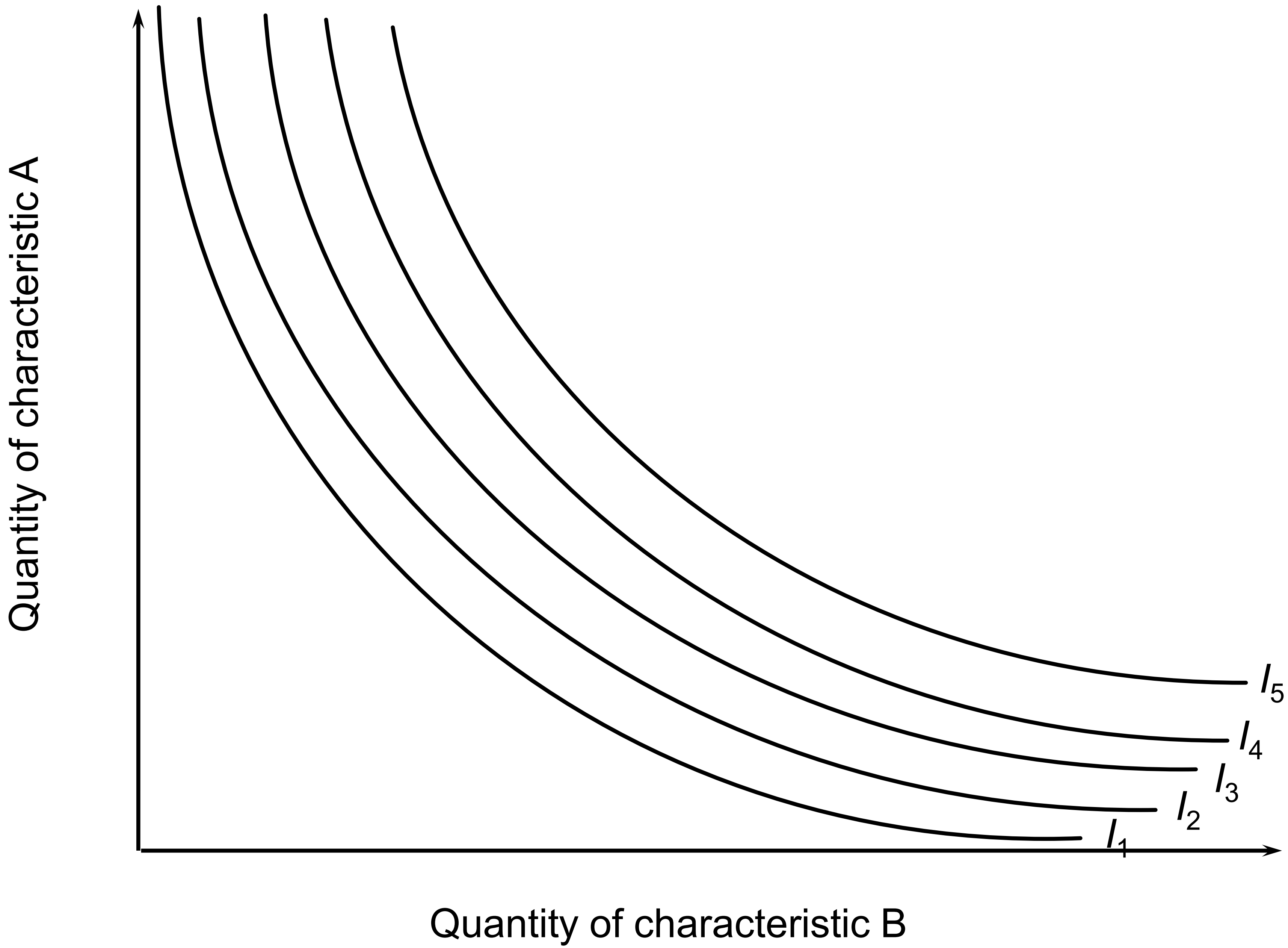
F. become flatter, passing through point x_1.



The Optimum Level of Consumption

- While the efficiency frontier shows what a consumer can afford, Indifference Curves show what the consumer wants.
- What is an Indifference Curve?
    - It represents all combinations of two characteristics (e.g., fiber and sugar) that provide the same level of total satisfaction.
    - The consumer is "indifferent" between any two points on the same curve.
- The Indifference Map:
    - Multiple curves (I_1 to I_5) represent different levels of utility.
    - Curves further from the origin represent higher levels of utility ($I_5 > I_4 > I_3$, etc.). Consumers always aim for the highest possible curve.

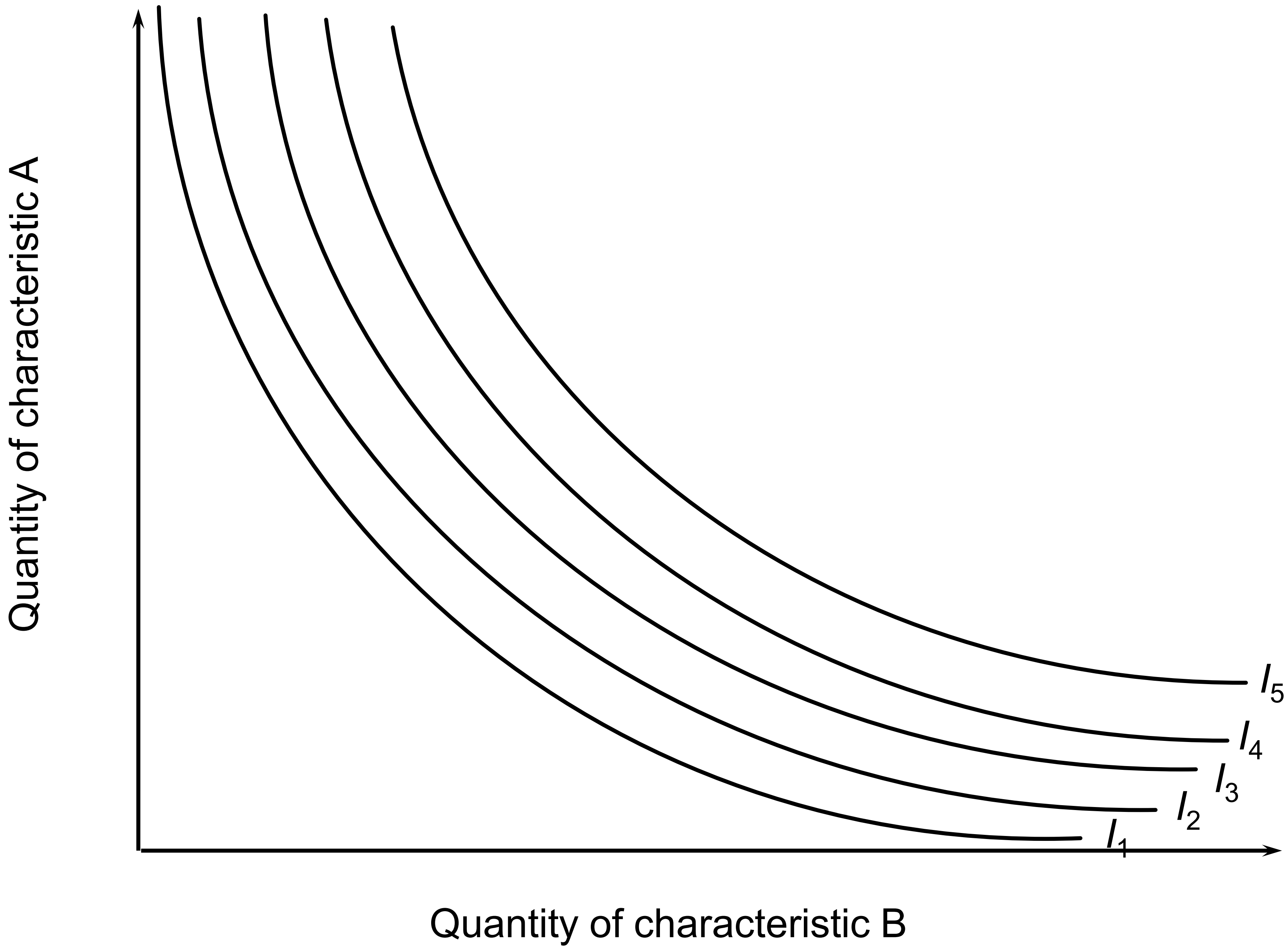
Choosing between brands



Shape and Variation of Indifference Curves

- **Downward Sloping:** To maintain the same level of utility, if a consumer loses some of one characteristic, they must receive more of the other to compensate.
- **Bowed Toward the Origin (Convex):**
  - This shape represents the Diminishing Marginal Rate of Substitution (MRS).
  - Consumers are willing to give up less and less of one attribute to get more of another as they already have a lot of the second attribute.
  - Example: If your phone has very little battery left but a very bright screen, you'd give up a lot of brightness for just a bit more battery life. If your battery is already full, you wouldn't give up much brightness to stretch it further.
- **Individual Differences**: Every consumer has a unique "indifference map." Some may prioritize fiber (health-conscious), while others prioritize sugar (taste-conscious), resulting in differently shaped curves.

Choosing between brands

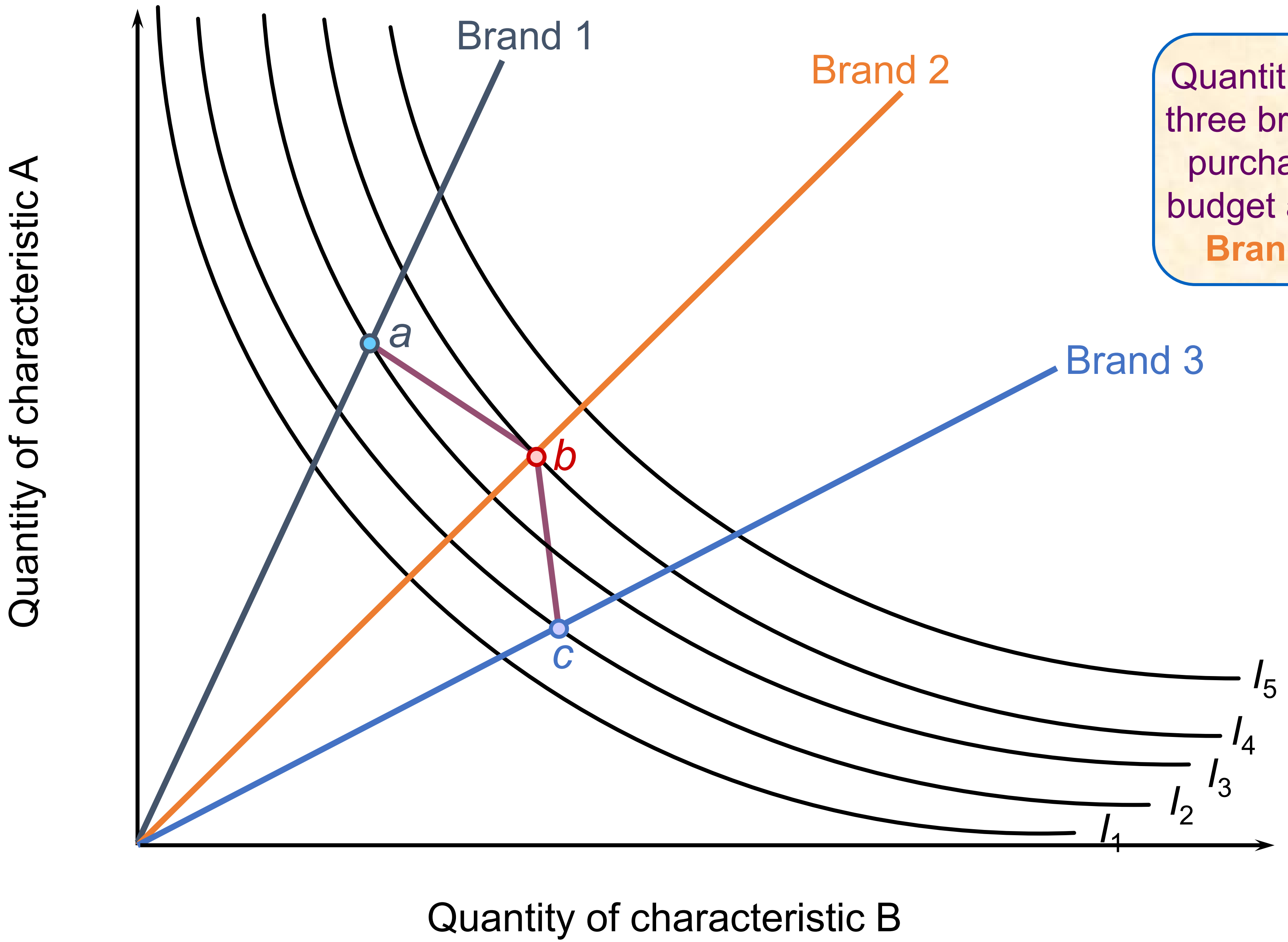


The Optimum Combination of Characteristics

- A "rational" consumer maximizes utility by reaching the highest indifference curve that touches their efficiency frontier.
- The consumer compares three brands (a, b, and c) on the efficiency frontier.
- Brand 2 (Point b): This point sits on the highest possible indifference curve.
- Brand 1 (Point a): Reaches a lower indifference curve.
- Brand 3 (Point c): Reaches the lowest curve of the three.

The Winner: The consumer will choose Brand 2 because it provides the specific mix of characteristics that aligns best with their personal preferences and budget.

Choosing between brands



Quantities of any one of three brands that can be purchased for a given budget at current prices. **Brand 2 is chosen**

Q A change in tastes between characteristics would be illustrated by a:

A. parallel shift in the indifference curves.

B. parallel shift in the efficiency frontier.

C. change in the shape of the indifference curves.

D. change in the slope of the efficiency frontier.

E. change in the slope of the product’s ray.

Business Strategy and the Characteristics Approach

- Firms must compete on more than just price; they must optimize product specifications (features) relative to rivals to capture consumer demand.
- Strategic Repositioning: A firm can swing its "brand ray" by changing its characteristic mix.
  - Example: Adding more of Characteristic B moves the ray clockwise.
  - The Substitute Effect: As a product’s features become more similar to a rival's, demand becomes more elastic because the products are now closer substitutes.

Business Strategy and the Characteristics Approach

- Influencing Consumer Tastes: Firms use marketing to shift the shape of indifference curves.
- If consumers are persuaded to value Characteristic A more, their indifference curves become shallower.
- This can move a firm’s current position (Point A) onto a higher indifference curve than a rival's (Point B), triggering a switch in consumer choice.
- New Product Launches: Firms use this analysis to find "gaps" in the market—positioning their ray and price to ensure they can pull sales away from existing competitors.

Limitations of Characteristics Analysis

- Measurement Difficulty: Not all attributes are "objective." Subjective features like design, style, or visual appeal depend on personal taste and are nearly impossible to quantify mathematically.
- The Complexity of Multi-Attributes: While the theory is easiest with two variables (one for each axis), real-world products have dozens of features. Adding more variables makes the mathematical modeling significantly more complex.

Limitations of Characteristics Analysis

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- Practicality of Indifference Curves:
    - Consumers struggle to accurately describe their own indifference points.
    - Aggregating the preferences of millions of diverse consumers into a single map is virtually impossible.
- Predicting Taste Volatility: Consumer preferences are not static. Changes in fashion or social trends shift the shape of indifference curves in ways that are extremely difficult for firms to forecast.

While not a perfect predictive tool, characteristics analysis provides a vital rough guide for market research and strategic planning.

Questions